

Analysis and Synthesis of Algorithms

Design of Algorithms

Linear Programming

The Simplex Method

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Outline

- Motivational Problem
- Canonical Forms: Standard and Slack
- Problem Formulation
- The Simplex Algorithm
- Feasible Initial Solutions
- Duality and Correctness of Simplex

Example: Procurement of Web Servers

- Suppose that a Web Server Company wants to buy new servers to replace outdated ones
- Models on the Market:
 - A standard model which costs \$400, uses 300W of power, takes up two shelves of a server rack, and can handle 1,000 hits/min.
 - A cutting-edge model, which costs \$1,600, uses 500W of power, but takes up only one shelf, and can handle 2,000 hits/min.
- **Your Mission:** Maximize the “Bang for the Buck”
 - Budget of \$36,800, 44 shelves of server space and 12,200W of power

Question:

How many units of each model should the company purchase in order to maximize the number of hits it can serve per minute?

Example: Formulation

- Variables:

x_1 = standard server; x_2 = spiffy server;

- Maximize:

$$\text{NumHits/Min} = 1000 x_1 + 2000 x_2$$

- Subject to Constraints:

$$\text{Budget: } 400 x_1 + 1600 x_2 \leq 36800$$

$$\text{Shelves: } 2 x_1 + x_2 \leq 44$$

$$\text{Power: } 300 x_1 + 500 x_2 \leq 12200$$

$$x_1, x_2 \geq 0$$

Example: Formulation

- Caveat:
 - In reality x_1 and x_2 need to be integer values...
- The “real formulation” calls for a ILP formulation
 - Harder to Solve Optimally
 - For now, we stick with this LP formulation...

$$\text{Maximize } c = 1000 x_1 + 2000 x_2$$

$$400 x_1 + 1600 x_2 \leq 36800$$

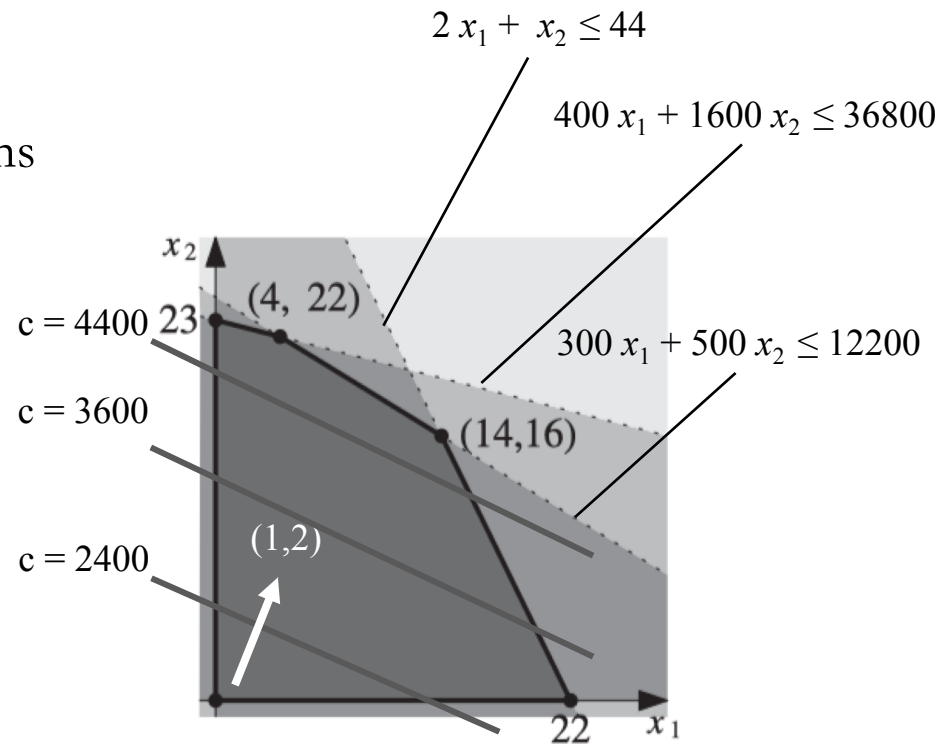
$$2 x_1 + x_2 \leq 44$$

$$300 x_1 + 500 x_2 \leq 12200$$

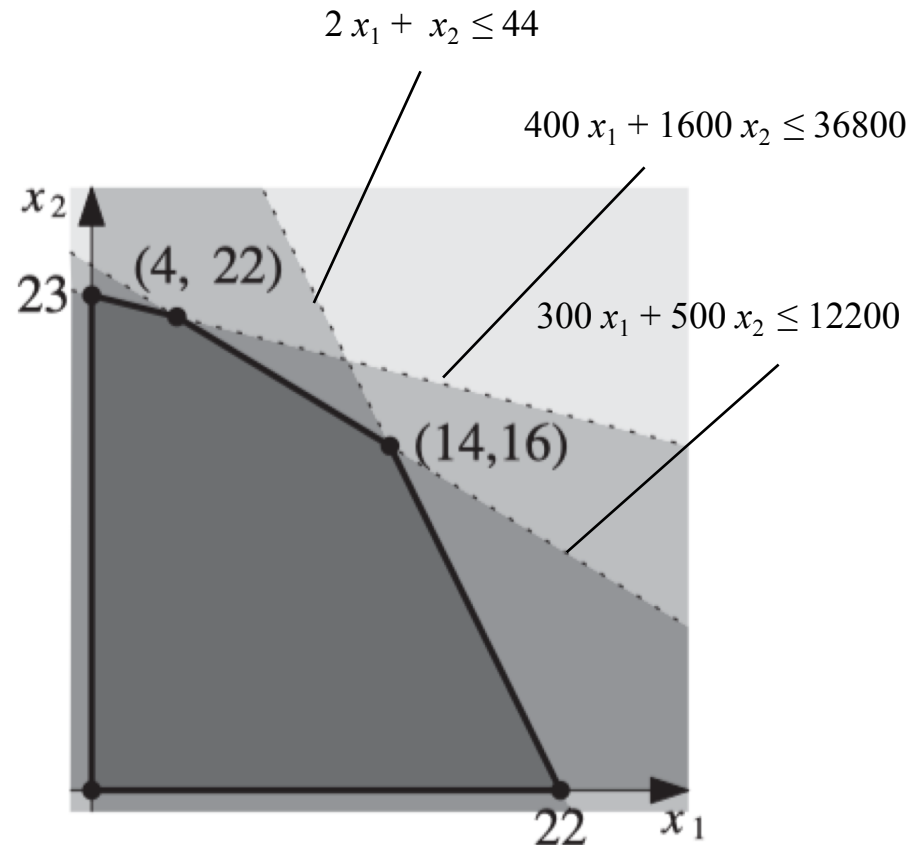
$$x_1, x_2 \geq 0$$

Example: Geometric View

- Domain: N-dim Space
 - Axis define values of variables $x_1, x_2 \geq 0$
- Constraints:
 - Space of Feasible Solutions
 - Convex Region
- Objective Function:
 - Defines Iso-Cost “plane”

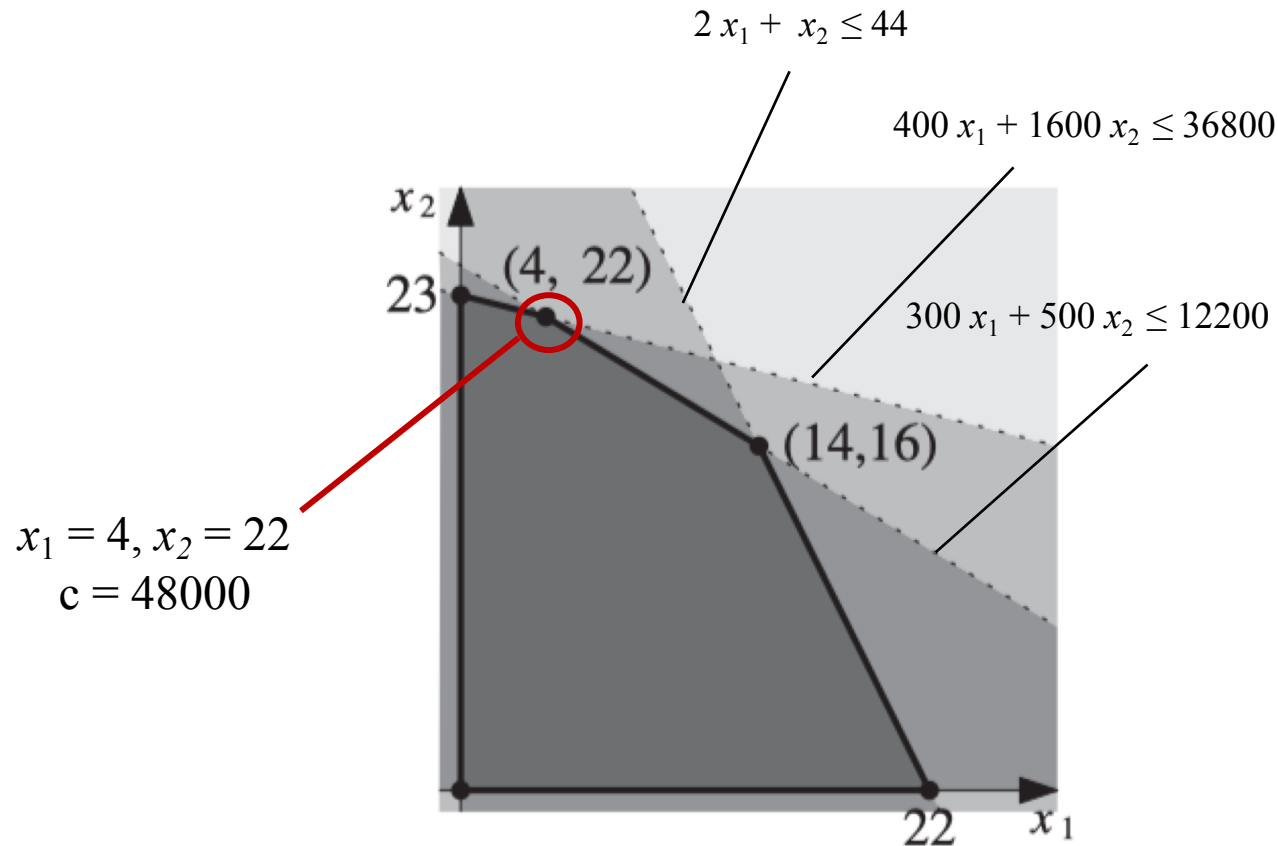


Example: Geometric View



- Optimal Solution
 - $x_1 = 4, x_2 = 22$ and value $c = 48000$

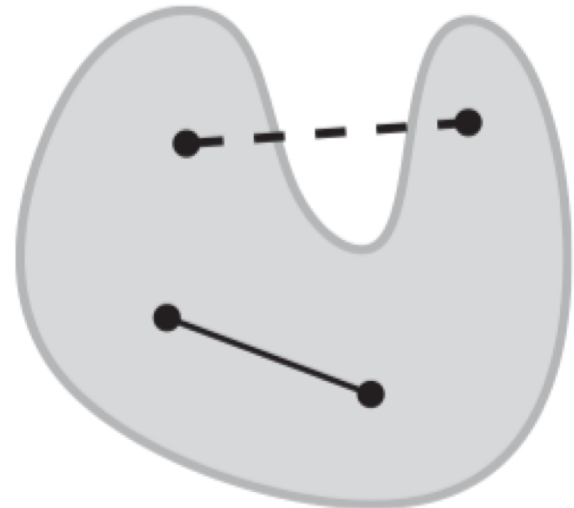
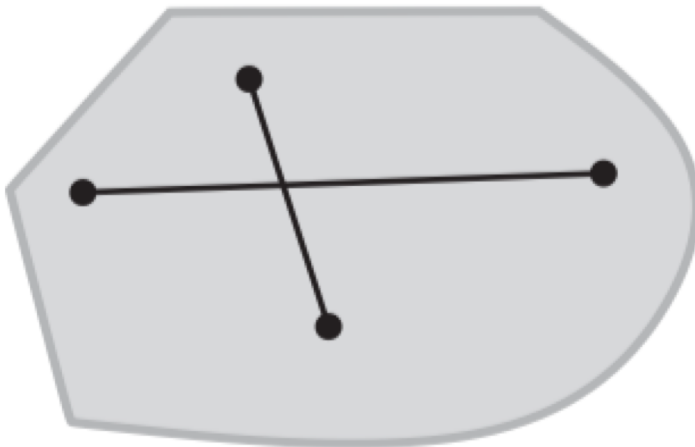
Example: Geometric View



- Optimal Solution
 - $x_1 = 4, x_2 = 22$ and value $c = 48000$

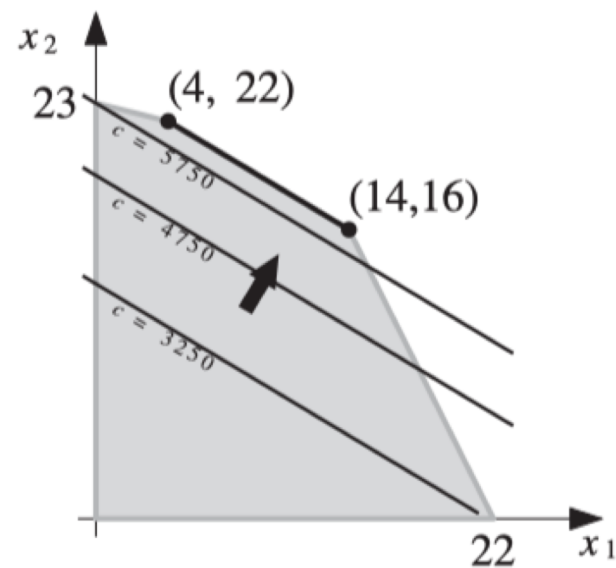
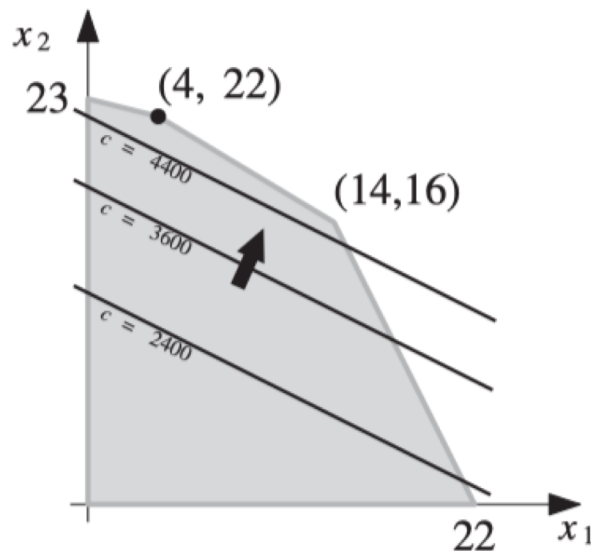
Convex Region

- Geometric Property
 - All points connecting any 2 points inside the region along a $N-1$ dimensional plane, are also inside the region.
- Important in Various Other Problems
 - Convex Hull in Geometric Problems



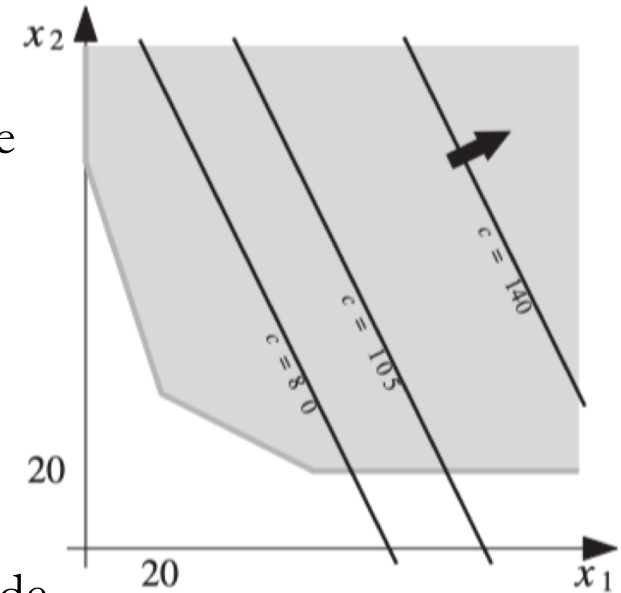
Facts on Uniqueness and Optimality

- **Fact:** Feasible Region is a Convex Polytope
 - $\Rightarrow$ Optimal solution, if exists, always occurs at the boundary
 - $\Rightarrow$ Optimal solution, if exists, **may not** be Unique



Existence of Optimal Solution

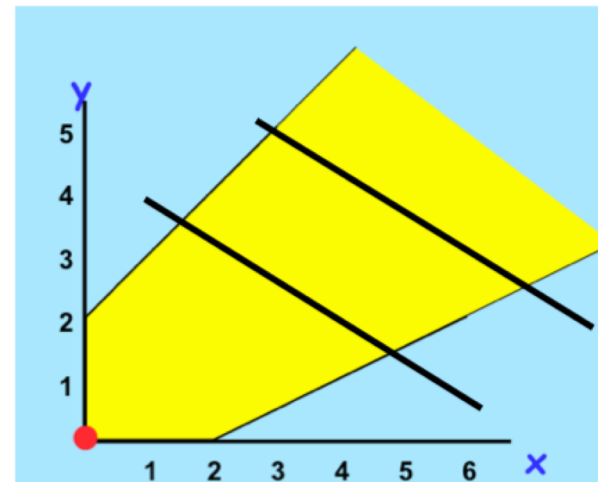
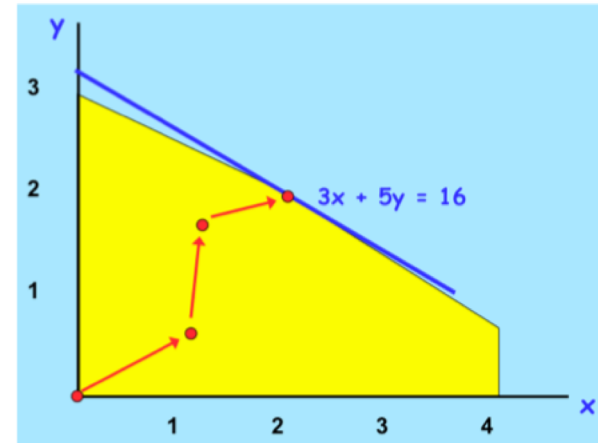
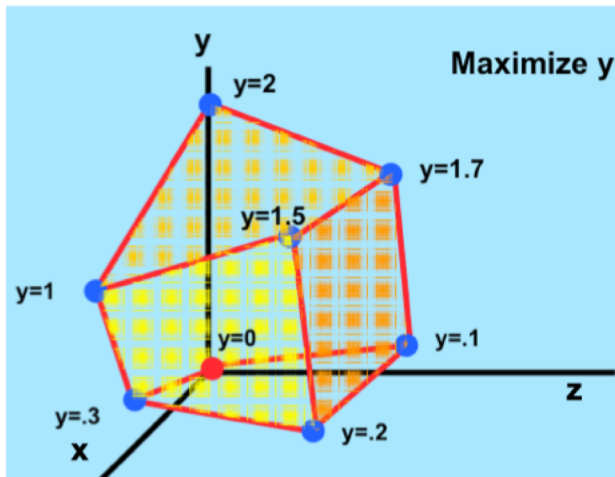
- A Optimal Solution may not exist
- Convex Region is Empty
 - Constraints are so restrictive that **no assignment** to the variables **can satisfy** every constraint.
 - Geometrically, the **intersection** of all the half-spaces **is empty**.
- Convex Region is Unbounded
 - Constraints do not define a bounded Convex Region
 - Unbounded Region $\Rightarrow$ We can move inside it and always find a “better solution” with higher value, Objective function tends to $+\infty$



Linear Programming Overview

- Any solution to the set of constraints is called a **feasible solution**
 - To each feasible solution corresponds a value of the **objective function** (or cost)
- Set of feasible solutions is denoted by **feasible region**
- Feasible region is a **Convex Region** in a N-dim space
 - S is Convex:
 - Any segment connecting two points in S never goes outside S
 - S is designated by **Simplex**

Feasible Region



General Formulation

- Linear Programming (LP):
 - Optimize (either minimize or maximize) a linear function subject to a set of linear constraints
 - Linear Objective Function:

$$f(x_1, x_2, \dots, x_n) = \sum_{j=1}^n c_j x_j$$

- Linear Constraints:

$$g_i(x_1, x_2, \dots, x_n) = \sum_{j=1}^n a_{ij} x_j \begin{matrix} \geq \\ = \\ \leq \end{matrix} b_i$$

A Simple Algorithm

- If the Feasible Region is Bounded*
 - There is a solution **and** it occurs at a Vertex or Verteces (**why?**)
- **Algorithm:**
 - Compute the Location of all the vertices,
 - Calculate the objective function at each of them
 - Find the smallest/largest value.
- **Issue:** Valid Approach but may not be Efficient
 - There might be an exponential number of vertices
- **Solution:** Smatter “Vertex-hunting” Approach
 - The Simplex Algorithm...

*At least if the value of the objective function is bounded below in the case of a minimization problem.

Linear Programming Overview

- Canonical LP Representation Forms:
 - **Standard** and **Slack** Forms
- Algorithms:
 - **Simplex Algorithm**
 - Follows a path through the vertices of the feasible region that increases the objective function in a **greedy** fashion;
 - Exponential in worst-case; very efficient in practice;
 - Widely used
 - Ellipsoid Algorithm
 - Polynomial; Often inefficient
 - Interior Point Methods
 - Polynomial; efficient in practice & competitive with the Simplex Algorithm

LP Overview: Basic Concepts

- Concepts:
 - Feasible Solution
 - Infeasible Solution
 - Objective Function Value
 - Maximum/minimum value: optimal objective value
 - An LP without feasible solutions is infeasible; otherwise is feasible
 - A feasible LP without optimal solution is unbounded
 - We can always find another point inside the Convex Region that is better than the previously found “best” point...
 - Convex region must be unbounded...

Standard Form

Maximize $\sum_{j=1}^n c_j x_j$ Objective Function

subject to $\sum_{j=1}^n a_{ij} x_j \leq b_i \quad i = 1, \dots, m$

Constraints $x_j \geq 0 \quad j = 1, \dots, n$

- Values c_j, a_{ij}, b_i are integers
- Matrix Representation: maximize $c^T x$
subject to $Ax \leq b$
 $x \geq 0$

– where $A = (a_{ij})$, $b = (b_i)$ and $c = (c_j)$ and $x = (x_j)$

Conversion to Standard Form

- Problems:
 - Minimizing instead of Maximizing
 - Variables without Constraints
 - Equality Constraints
 - Greater Than Constraints ($\geq$)

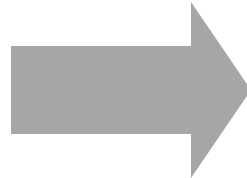
Conversion to Standard Form

- Solutions:
 - Minimizing instead of Maximizing:
 - Multiply Coefficients of Objective Function by -1
 - Variables without Constraints :
 - Replace x_i by 2 variables x_{i1} and x_{i2} , and multiply coefficients of x_{i2} by -1 (if x_i is positive, x_{i1} is positive and x_{i2} is zero, for instance...)
 - Both variables will have a $\leq$ constraints,
 - One of them has a -1 coefficient.
 - Equality Constraints:
 - Include two constraints, one with $\geq$, another with $\leq$
 - Greater Than Constraints $\geq$
 - Multiply associated coefficients in constraint by -1

Conversion to Standard Form

$$\max \quad 2x_1 - 3x_2$$

$$\begin{aligned} x_1 + x_2 &= 7 \\ x_1 - 2x_2 &\leq 4 \\ x_1 &\geq 0 \end{aligned}$$



$$\max \quad 2x_1 - 3x'_2 + 3x''_2$$

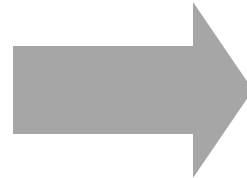
$$\begin{aligned} x_1 + x'_2 - x''_2 &= 7 \\ x_1 - 2x'_2 + 2x''_2 &\leq 4 \\ x_1, x'_2, x''_2 &\geq 0 \end{aligned}$$

no constraint on x_2 ,
then add x'_2 and x''_2

Conversion to Standard Form

$$\max \quad 2x_1 - 3x_2$$

$$\begin{aligned} x_1 + x_2 &= 7 \\ x_1 - 2x_2 &\leq 4 \\ x_1 &\geq 0 \end{aligned}$$



$$\max \quad 2x_1 - 3x'_2 + 3x''_2$$

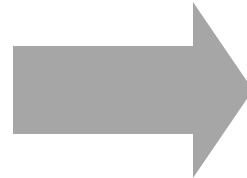
$$\begin{aligned} x_1 + x'_2 - x''_2 &\leq 7 \\ x_1 + x'_2 - x''_2 &\geq 7 \\ x_1 - 2x'_2 + 2x''_2 &\leq 4 \\ x_1, x'_2, x''_2 &\geq 0 \end{aligned}$$

equality constraint on
first equation

Conversion to Standard Form

$$\max \quad 2x_1 - 3x_2$$

$$\begin{aligned} x_1 + x_2 &= 7 \\ x_1 - 2x_2 &\leq 4 \\ x_1 &\geq 0 \end{aligned}$$



$$\max \quad 2x_1 - 3x'_2 + 3x''_2$$

$$\begin{aligned} x_1 + x'_2 - x''_2 &\leq 7 \\ -x_1 - x'_2 + x''_2 &\leq -7 \\ x_1 - 2x'_2 + 2x''_2 &\leq 4 \\ x_1, x'_2, x''_2 &\geq 0 \end{aligned}$$

normalization on
inequality

Conversion to Slack Form

- Slack Form works with Equality Constraints Only
 - Basis of the Simplex Algorithm
 - Replace all inequality constraints with equality constraints
- For each constraint, introduce an additional Slack variable s_i
 - This “slack” variable, tell you ”how far from the constraint” the original variable is
 - So, it is at the “boundary” when it is 0.

$$s_i = b_i - \sum_{j=1}^n a_{ij}x_j$$

$$s_i \geq 0$$

Conversion to Slack Form

- Expressions: $x_{n+1} = b_i - \sum_{j=1}^n a_{ij}x_j$
 $x_{n+1} \geq 0$
 - Variables expressed in terms of other variables are designated by **Basic Variables**
 - Variables that define the basic variables are called **Non-Basic Variables**
- Objective to Maximize: $z = \sum_{j=1}^n c_j x_j + v$
- Slack Form described as: (N, B, A, b, c, v)
 - N: Non-Basic Variables; $|N| = n$
 - B: Basic Variables; $|B| = m$
 - v: Objective Function Constant

Slack Form

$$\max \quad 2 x_1 - 3 x_2' + 3 x_2''$$

$$\begin{aligned} x_1 + x_2' - x_2'' &\leq 7 \\ -x_1 - x_2' + x_2'' &\leq -7 \\ x_1 - 2x_2' + 2x_2'' &\leq 4 \\ x_1, x_2', x_2'' &\geq 0 \end{aligned}$$

$$z = 2 x_1 - 3 x_2 + 3 x_3$$

$$\begin{aligned} x_4 &= 7 - x_1 - x_2 + x_3 \\ x_5 &= -7 + x_1 + x_2 - x_3 \\ x_6 &= 4 - x_1 + 2 x_2 - 2 x_3 \\ x_1, x_2, x_3, x_4, x_5, x_6 &\geq 0 \end{aligned}$$

Slack Form

$$\max \quad 2x_1 - 3x_2' + 3x_2''$$

$$z = 2x_1 - 3x_2 + 3x_3$$

$$\begin{aligned} x_1 + x_2' - x_2'' &\leq 7 \\ -x_1 - x_2' + x_2'' &\leq -7 \\ x_1 - 2x_2' + 2x_2'' &\leq 4 \\ x_1, x_2', x_2'' &\geq 0 \end{aligned}$$

$$\begin{aligned} x_4 &= 7 - x_1 - x_2 + x_3 \\ x_5 &= -7 + x_1 + x_2 - x_3 \\ x_6 &= 4 - x_1 + 2x_2 - 2x_3 \\ x_1, x_2, x_3, x_4, x_5, x_6 &\geq 0 \end{aligned}$$

Basic Variables

Non-Basic Variables

The Simplex Algorithm

- LP Problem in Slack Form
- Initial Feasible Solution
- Iteratively Pivots to Improve Objective Function
- Greedy choice of Pivot Variables

- Algorithm Outline:
 1. Find a Vertex inside the Feasible region.
 2. While there is a neighbor x of the current vertex with a larger value of $c \cdot x$, move to it.
 3. Terminate when you reach an optimum (whenever there are no better neighboring vertices)

Pivoting: Moving Across Vertices

- What Does it Mean to be at a Vextex?
 - It means that the set of non-basic variables are all 0.0
 - The value of the Cost function is given by value of v

$$z = \sum_{j=1}^n c_j x_j + v$$

- How to Move to a new Vextex?
 - This means that we must take a Variable that is currently zero (Non-Basic) and make it non-zero (Basic).
 - The variable which is replaced (in the basic set) is called the leaving variable (goes from the basic to the non-basic set)
 - The variable which replaces it, is known as the entering variable (goes from being the non-basic to the basic).

Simplex Iteration and Pivots

- Initial Basic Solution
 - Set Non-Basic Variables to 0.0, Basic Variables to b_i
 - Feasible? Why would it not be feasible? What to do? (**later**)
- Choose a Pivot Non-Basic variable $\mathbf{x}_e$ such that
 - Value of Objective function Increases (the most)
 - Bounded by a Basic Variable Constraints (positivity)
 - Non-basic variable $\mathbf{x}_e$ will become a basic variable

$$\text{Entry Variable} \quad N' = N - \{X_e\} \cup \{X_l\}$$

- Choose Basic variable $\mathbf{x}_l$ to become a Non-Basic variable

$$\text{Exiting variable} \quad B' = B - \{X_l\} \cup \{X_e\}$$

- Determine new slack form (N', B', A, b, c, v)

The Simplex Algorithm

- Determine Initial Slack Form
 - For a Feasible Initial Solution
 - Otherwise the problem is Infeasible, and the algorithm returns “unfeasible” and terminates.

$$z = \sum_{j=1}^n c_j x_j + v$$

- While there Exists $c_{entry} > 0$ (i.e., z value will increase)
 - x_{entry} defines a new entry variable (i.e., new basic variable)
 - Select x_{exit}
 - x_{exit} corresponds to a constraint i that **minimizes** b_i / a_i , for $a_i < 0$
 - If $a_i > 0$ for all i , then the problem is ‘unbounded’
 - Pivoting with (N, B, A, b, c, v, *exit*, *entry*)

Simplex Algorithm Outline

1. start with a problem in Slack Form

2. repeat

- Choose a variable x_i with positive coefficient in the objective function.
- Find the equation $x = b + c x_i + \dots$ where $c < 0$ and $-b/c$ is minimal
- Rewrite this equation with x_i the subject $x_i = -b/c + 1/c x + \dots$
- Substitute $-b/c + 1/c x + \dots$ for x_i in all other equations and objective function

until no such variable x_i exists or no such equation exists

3. if no such x_i exists then

Optimum is found

else

there is no Optimal solution

Simplex Algorithm Outline

1. start with a problem in Slack Form

2. repeat

- Choose a variable x_i with positive coefficient in the objective function.
- Find the equation $x = b + c x_i + \dots$ where $c < 0$ and $-b/c$ is minimal
- Rewrite this equation with x_i the subject $x_i = -b/c + 1/c x + \dots$
- Substitute $-b/c + 1/c x + \dots$ for x_i in all other equations and objective function

until no such variable x_i exists or no such equation exists

3. if no such x_i exists then

Optimum is found

else

there is no Optimal solution

Simplex Algorithm - Example

$$\max 3 x_1 + 1 x_2 + 2 x_3$$

$$z = 3 x_1 + 1 x_2 + 2 x_3$$

$$x_1 + x_2 + 3 x_3 \leq 30$$

$$2 x_1 + 2 x_2 + 5 x_3 \leq 24$$

$$4 x_1 + x_2 + 2 x_3 \leq 36$$

$$x_1, x_2, x_3 \geq 0$$

$$x_4 = 30 - 1 x_1 - 1 x_2 - 3 x_3$$

$$x_5 = 24 - 2 x_1 - 2 x_2 - 5 x_3$$

$$x_6 = 36 - 4 x_1 - 1 x_2 - 2 x_3$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

Standard Form

Slack Form

Simplex Algorithm - Example

$$z = 3 x_1 + 1 x_2 + 2 x_3$$

The Initial Basic Feasible Solution is (0,0,0,30,24,36) with objective value zero.

$$x_4 = 30 - 1 x_1 - 1 x_2 - 3 x_3$$

$$x_5 = 24 - 2 x_1 - 2 x_2 - 5 x_3$$

$$x_6 = 36 - 4 x_1 - 1 x_2 - 2 x_3$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

We now choose x_1 as the entering variable and x_6 as the leaving variable - **why?**

Simplex Algorithm - Example

$$z = 3x_1 + 1x_2 + 2x_3$$

$$x_4 = 30 - 1x_1 - 1x_2 - 3x_3$$

$$x_5 = 24 - 2x_1 - 2x_2 - 5x_3$$

$$x_6 = 36 - 4x_1 - 1x_2 - 2x_3$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

x_1 has the largest coeff. in z so an increase in x_1 will lead to the biggest increase in z

How much can we increase x_1 ?

Dictated by constraints as

x_4, x_5, x_6 cannot become negative

Compare:

$30/1$ (for x_4),

$24/2$ (for x_5),

$36/4$ (for x_6)

We now choose x_1 as the entering variable and x_6 as the leaving variable - **why?**

Simplex Algorithm - Example

$$z = 3x_1 + 1x_2 + 2x_3$$

$$x_4 = 30 - 1x_1 - 1x_2 - 3x_3$$

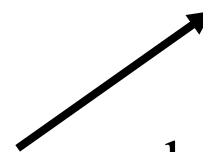
$$x_5 = 24 - 2x_1 - 2x_2 - 5x_3$$

$$x_6 = 36 - 4x_1 - 1x_2 - 2x_3$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

Rewrite x_1 as a function of x_6

$$x_1 = 9 - \frac{x_2}{4} - \frac{x_3}{2} - \frac{x_6}{4}$$



replace in other equality constraints

and

in Cost value function

Simplex Algorithm - Example

$$z = 27 + \frac{x_2}{4} + \frac{x_3}{2} - \frac{3x_6}{4}$$

$$x_1 = 9 - \frac{x_2}{4} - \frac{x_3}{2} - \frac{3x_6}{4}$$

$$x_4 = 21 - \frac{3x_2}{4} - \frac{5x_3}{2} + \frac{x_6}{4}$$

$$x_5 = 6 - \frac{3x_2}{2} - \frac{4x_3}{1} + \frac{x_6}{2}$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

The Basic Feasible Solution is now (9,0,0,21,6,0) with objective value 27.

Simplex Algorithm - Example

$$z = 27 + \frac{x_2}{4} + \frac{x_3}{2} - \frac{3x_6}{4}$$

$$x_1 = 9 - \frac{x_2}{4} - \frac{x_3}{2} - \frac{3x_6}{4}$$

$$x_4 = 21 - \frac{3x_2}{4} - \frac{5x_3}{2} + \frac{x_6}{4}$$

$$x_5 = 6 - \frac{3x_2}{2} - \frac{4x_3}{1} + \frac{x_6}{2}$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

$$z = \frac{111}{4} + \frac{x_2}{16} - \frac{x_5}{8} - \frac{11x_6}{16}$$

$$x_1 = \frac{33}{4} - \frac{x_2}{16} - \frac{x_5}{8} - \frac{11x_6}{16}$$

$$x_3 = \frac{3}{2} - \frac{3x_2}{8} - \frac{x_5}{4} + \frac{x_6}{8}$$

$$x_4 = \frac{69}{4} + \frac{3x_2}{16} + \frac{5x_5}{8} - \frac{x_6}{16}$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

We now choose x_3 as the entering variable
and x_5 as the leaving variable - **why?**

Simplex Algorithm - Example

$$z = \frac{111}{4} + \frac{x_2}{16} - \frac{x_5}{8} - \frac{11x_6}{16}$$

$$x_1 = \frac{33}{4} - \frac{x_2}{16} - \frac{x_5}{8} - \frac{11x_6}{16}$$

$$x_3 = \frac{3}{2} - \frac{3x_2}{8} - \frac{x_5}{4} + \frac{x_6}{8}$$

$$x_4 = \frac{69}{4} + \frac{3x_2}{16} + \frac{5x_5}{8} - \frac{x_6}{16}$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

$$z = 28 - \frac{x_3}{6} - \frac{x_5}{6} - \frac{2x_6}{3}$$

$$x_1 = 8 + \frac{x_3}{6} + \frac{x_5}{6} - \frac{x_6}{3}$$

$$x_2 = 4 - \frac{8x_3}{3} - \frac{2x_5}{3} + \frac{x_6}{3}$$

$$x_4 = 18 - \frac{x_3}{2} + \frac{x_5}{2}$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

We now choose x_2 as the entering variable
and x_3 as the leaving variable - **why?**

Simplex Algorithm - Example

$$z = 28 - \frac{x_3}{6} - \frac{x_5}{6} - \frac{2x_6}{3}$$

$$x_1 = 8 + \frac{x_3}{6} + \frac{x_5}{6} - \frac{x_6}{3}$$

$$x_2 = 4 - \frac{8x_3}{3} - \frac{2x_5}{3} + \frac{x_6}{3}$$

$$x_4 = 18 - \frac{x_3}{2} + \frac{x_5}{2}$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

The basic feasible solution is now (8,4,0,18,0,0) with objective value 28.

Now, all Non-Basic Variables have negative coefficients in z so we really cannot improve on z anymore (an increase in any of x_1 , x_5 or x_6 would decrease z).

The Optimal Solution to our problem:

$$x_1 = 8$$

$$x_2 = 4$$

$$x_4 = 18$$

and $z = 28$

Initial Feasible Solution

- A Linear Programming Problem may be Feasible but the Initial Basic Solution may not be Feasible at all...

$$\text{maximize } z = 2x_1 - x_2$$

$$x_3 = 2 - 2x_1 + x_2$$

$$x_4 = -4 - x_1 + 5x_2$$

$$x_1, x_2, x_3, x_4 \geq 0$$

The basic solution

$(0, 0, 2, -4)$ is infeasible!

Original Problem L
(Slack Form)

Initial Feasible Solution

- A Linear Programming Problem may be Feasible but the Initial Basic Solution may not be Feasible at all...

$$\begin{array}{ll} \text{maximize} & z = 2x_1 - x_2 \\ & x_1 = 0 \\ & x_2 = 0 \end{array}$$

$$x_3 = 2 - 2x_1 + x_2$$

$$x_4 = -4 - x_1 + 5x_2$$

$$x_1, x_2, x_3, x_4 \geq 0$$

Original Problem L
(Slack Form)

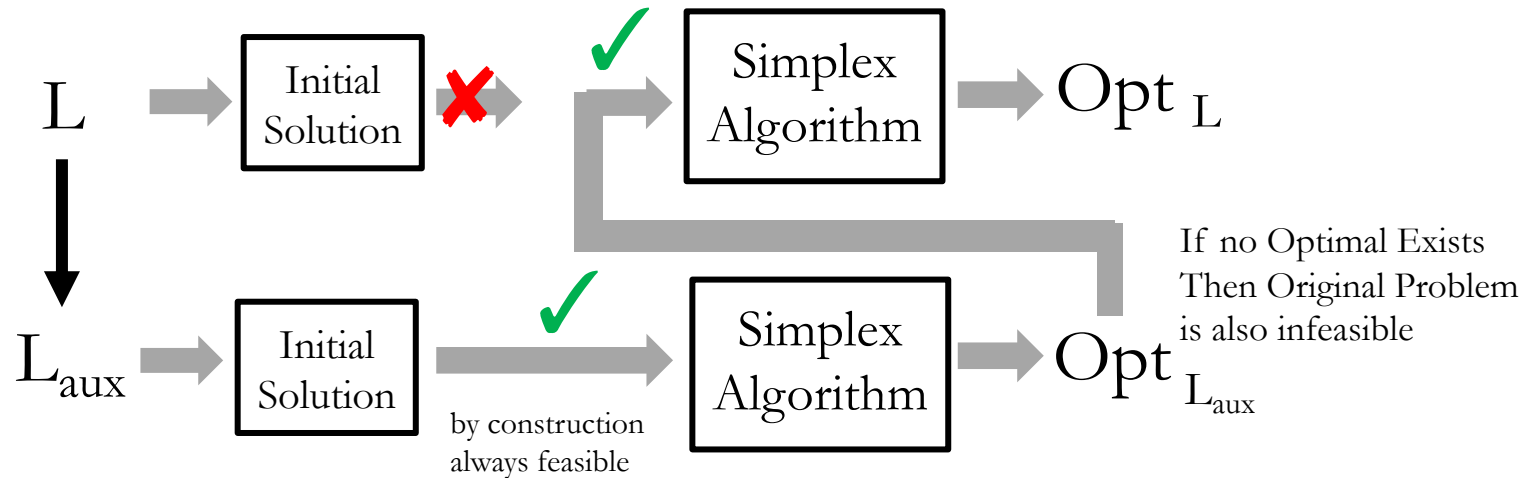
The basic solution
(0, 0, 2, -4) is infeasible!

$$x_4 = -4 \leq 0$$

$$x_4 \geq 0$$

Initial Feasible Solution

- A Linear Programming Problem may be Feasible but the Initial Basic Solution may not be Feasible at all...
- **Solution:**
 - Create an Auxiliary Linear Programming Problem;
 - Solve it and then revisit the original problem with the solution to the auxiliary problem as the new initial solution.



Initial Feasible Solution

- Let L be an LP Problem in Standard Form, and let L_{aux} be defined as follows:

$$\begin{array}{ll}
 \text{maximize} & -x_0 \\
 \text{subject to} & \sum_{j=1}^n a_{ij}x_j - x_0 \leq b_i \quad i = 1, 2, \dots, m \\
 & x_j \geq 0 \quad j = 0, 1, 2, \dots, n
 \end{array}$$

new variable...

$$\begin{array}{ll}
 \text{maximize} & 2x_1 - x_2 \\
 & 2x_1 - x_2 \leq 2 \\
 & x_1 - 5x_2 \leq -4 \\
 & x_1, x_2 \geq 0
 \end{array}$$

Original Problem L
(standard form)

$$\begin{array}{ll}
 \text{maximize} & -x_0 \\
 & 2x_1 - x_2 - x_0 \leq 2 \\
 & x_1 - 5x_2 - x_0 \leq -4 \\
 & x_0, x_1, x_2 \geq 0
 \end{array}$$

Auxiliary Problem L_{aux}
(standard form)

Initial Feasible Solution

- Let L be an LP Problem in Standard Form, and let L_{aux} be defined as follows:

$$\begin{array}{ll} \text{maximize} & -x_0 \\ \text{subject to} & \sum_{j=1}^n a_{ij}x_j - x_0 \leq b_i \quad i = 1, 2, \dots, m \\ & x_j \geq 0 \quad j = 0, 1, 2, \dots, n \end{array}$$
- Then, L is feasible iff the optimal objective value of L_{aux} is 0
 - If L has solution, then L_{aux} has solution with $x_0 = 0$, the optimal value
 - If the optimal value of x_0 is 0, then the solution to L_{aux} is also a solution to L

Initial Feasible Solution

- If the Basic Initial Solution is Infeasible:
 - From L construct L_{aux}
 - But L_{aux} may also have an infeasible initial solution...
- Solution:
 - Pivot with *entry* = 0 (i.e., with x_0)
 - Determine *exit* to be the index i with the lowest b_i
 - The basic solution from this Pivot is feasible for L_{aux}
- Next Step:
 - Use Simplex to compute Optimal Solution of L_{aux}
 - If optimal solution of L_{aux} verifies $x_0 = 0$, use optimal solution of L_{aux} as initial solution of original LP problem, without x_0
 - Otherwise, original L is infeasible (Proof: CLRS 2nd Ed. Page 812)

Problem Formulations

$$\text{maximize } 2x_1 - x_2$$

$$2x_1 - x_2 \leq 2$$

$$x_1 - 5x_2 \leq -4$$

$$x_1, x_2 \geq 0$$

Original Problem L
(standard form)

$$\text{maximize } -x_0$$

$$2x_1 - x_2 - x_0 \leq 2$$

$$x_1 - 5x_2 - x_0 \leq -4$$

$$x_1, x_2, x_0 \geq 0$$

Auxiliary Problem L_{aux}
(standard form)

$$\text{maximize } z = 2x_1 - x_2$$

$$x_3 = 2 - 2x_1 + x_2$$

$$x_4 = -4 - x_1 + 5x_2$$

$$x_1, x_2, x_3, x_4 \geq 0$$

Original Problem L
(slack form)

$$\text{maximize } z = -x_0$$

$$x_3 = 2 - 2x_1 + x_2 + x_0$$

$$x_4 = -4 - x_1 + 5x_2 + x_0$$

$$x_0, x_1, x_2, x_3, x_4 \geq 0$$

Auxiliary Problem L_{aux}
(slack form)

Initial Feasible Solution – First Pivot

- After the first Pivot the Solution is Feasible for L_{aux}
 - Set index of *entry* = 0 (i.e., x_0)
 - Find $i=\text{low}$ such that $b_{\text{low}} < b_i$, $i=1, \dots, m$ (minimum b_i)
 - $b_{\text{low}} < 0$, given that the initial solution is only feasible if $b_i \geq 0$
 - and that is why we had to do this first Pivot operation...
 - After Pivoting we have:

$$x_0 = b_{\text{low}} / a_{\text{low},0} \quad \text{and} \quad x_i = b_i - a_{i,0} (b_{\text{low}} / a_{\text{low},0}), \quad i \neq 0$$

given that $a_{i,0} = -1$ for all i , (the coefficients for x_0 are initially all -1 in the L_{aux} standard formulation)

$$x_0 = -b_{\text{low}} > 0 \quad \text{and} \quad x_i = b_i - b_{\text{low}} > 0$$
 - which is now a feasible initial solution for L_{aux}

Note: $a_{i,0}$ denotes the initial value (at the zero^{th} iteration) of the coefficient a_i in the standard (not the slack) auxiliary problem formulation

Initial Feasible Solution

$$\text{maximize } z = 2x_1 - x_2$$

$$x_3 = 2 - 2x_1 + x_2$$

$$x_4 = -4 - x_1 + 5x_2$$

$$x_1, x_2, x_3, x_4 \geq 0$$

Original Problem L
(slack form)

$$\text{maximize } z = -x_0$$

$$x_3 = 2 - 2x_1 + x_2 + x_0$$

$$x_4 = -4 - x_1 + 5x_2 + x_0$$

$$x_0, x_1, x_2, x_3, x_4 \geq 0$$

Auxiliary Problem L_{aux}
(slack form)

Initial Basic Solution still
infeasible so, we pivot

Initial Feasible Solution

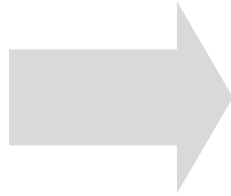
x_4 leaves and x_0 enters

$$z = -x_0$$

$$x_3 = 2 - 2x_1 + x_2 + x_0$$

$$x_4 = -4 - x_1 + 5x_2 + x_0$$

$$x_0, x_1, x_2, x_3, x_4 \geq 0$$



$$z = -4 - x_1 + 5x_2 - x_4$$

$$x_0 = 4 + x_1 - 5x_2 + x_4$$

$$x_3 = 6 - x_1 - 4x_2 + x_4$$

$$x_0, x_1, x_2, x_3, x_4 \geq 0$$

now the basic solution is
(4,0,0,6,0) which is feasible

Initial Feasible Solution

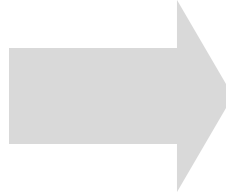
x_0 leaves and x_2 enters

$$z = -4 - x_1 + 5x_2 - x_4$$

$$x_0 = 4 + x_1 - 5x_2 + x_4$$

$$x_3 = 6 - x_1 - 4x_2 + x_4$$

$$x_0, x_1, x_2, x_3, x_4 \geq 0$$



$$z = -x_0$$

$$x_2 = \frac{4}{5} - \frac{x_0}{5} + \frac{x_1}{5} + \frac{x_4}{5}$$

$$x_3 = \frac{14}{5} + \frac{4x_0}{5} - \frac{9x_1}{5} + \frac{x_4}{5}$$

$$x_0, x_1, x_2, x_3, x_4 \geq 0$$

the new basic solution is $(0, 0, 4/5, 14/5, 0)$
which is feasible and is optimal as $x_0 = 0$

Initial Feasible Solution

$$\text{maximize } z = 2x_1 - x_2$$

$$x_3 = 2 - 2x_1 + x_2$$

$$x_4 = -4 - x_1 + 5x_2$$

$$x_1, x_2, x_3, x_4 \geq 0$$

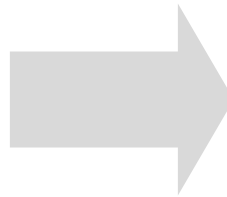
Original Problem L
(slack form)

$$z = -x_0$$

$$x_2 = \frac{4}{5} - \frac{x_0}{5} + \frac{x_1}{5} + \frac{x_4}{5}$$

$$x_3 = \frac{14}{5} + \frac{4x_0}{5} - \frac{9x_1}{5} + \frac{x_4}{5}$$

$$x_0, x_1, x_2, x_3, x_4 \geq 0$$



$$z = -\frac{4}{5} + \frac{9x_1}{5} - \frac{x_4}{5}$$

$$x_2 = \frac{4}{5} + \frac{x_1}{5} + \frac{x_4}{5}$$

$$x_3 = \frac{14}{5} - \frac{9x_1}{5} + \frac{x_4}{5}$$

$$x_1, x_2, x_3, x_4 \geq 0$$

Optimal Solution to
Auxiliary Problem L_{aux}

Original Problem L with new
basic feasible solution this time

Simplex Algorithm: Formal Results

- Given an LP Problem (A, b, c) :
 - If the Simplex Algorithm returns a Solution, then this Solution is a Feasible Solution
 - If the Simplex Algorithm returns “unbounded” then the Problem is Unbounded

- Given an LP Problem (A, b, c) in standard form, and B is a set of basic variables its Slack Form is unique

Simplex Algorithm: Formal Results

- Variation of Objective Function after pivoting:
 - Value of objective function does not decrease
 - Chosen variable has c_i positive
 - By the constraints definition the value of a variable is non-negative, so objective function will not decrease
 - Objective function value may not increase
 - Degenerative case
 - It is possible to ensure that Simplex Algorithm terminates

Simplex Algorithm: Formal Results

- The Simplex algorithm will **Cycle** if there exists identical Slack forms for two algorithm iterations
- If the Simplex algorithm does not terminate after $C_m^{n+m} = \binom{n+m}{m}$ iterations, then the algorithm is in a loop
 - Each B set determines uniquely the slack form
 - There are $n+m$ variables in $|B| = C_m^{n+m}$
 - Number of ways to choose B: C_m^{n+m}
 - Number of distinct slack forms: C_m^{n+m}
 - If algorithm executes more than C_m^{n+m} iterations, then it is in a loop
- How to Eliminate Cycles:
 - **Bland Rule:** chose variable with lower index in case of a tie

Duality in Linear Programming

- For Every Primal Problem there exists a corresponding Dual Problem:
 - Primal Maximization $\rightarrow$ Dual Minimization
 - Primal Minimization $\rightarrow$ Dual Maximization

Primal Problem:

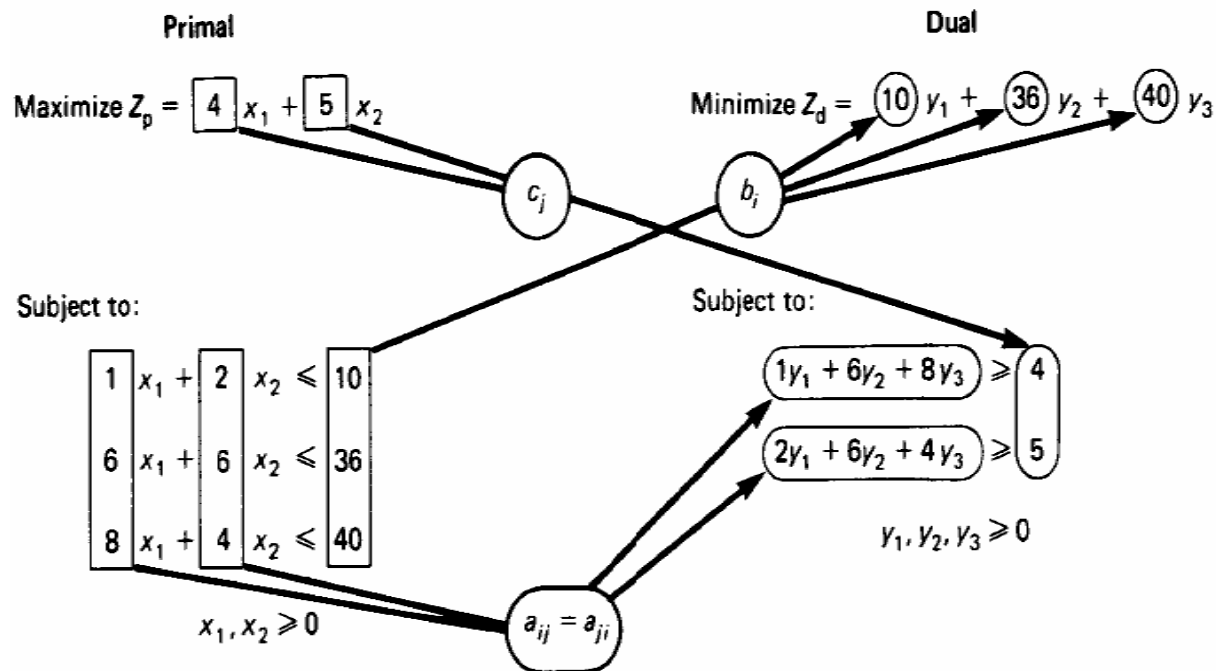
$$\begin{aligned}
 &\text{Maximize} && \sum_{j=1}^n c_j x_j \\
 &\text{subject to} && \sum_{j=1}^n a_{ij} x_j \leq b_i \quad i = 1, \dots, m \\
 & && x_j \geq 0 \quad j = 1, \dots, n
 \end{aligned}$$

Dual LP Problem:

$$\begin{aligned}
 &\text{Minimize} && \sum_{i=1}^m b_i y_i \\
 &\text{subject to} && \sum_{i=1}^m a_{ij} y_i \geq c_j \quad j = 1, 2, \dots, n \\
 & && y_i \geq 0 \quad i = 1, 1, 2, \dots, m
 \end{aligned}$$

Duality in Linear Programming

- Primal has n choice variable and m constraints and dual has m variables and n constraints
- Right hand side elements (b_i) in the primal correspond to coefficient of the objective function of the dual
- The a_{ij} constraint coefficients become a_{ji} in dual



Weak Duality in Linear Programming

- Let x be any Feasible Solution of the Primal Problem and let y be one Feasible Solution of the Dual Problem
- Under these conditions the following holds:

$$\sum_{j=1}^n c_j x_j \leq \sum_{i=1}^m b_i y_i$$

Duality in Linear Programming

- Let x be any solution derived by the Simplex Algorithm and let N and B be the set of variables in the final Slack Form.
- Let c be the vector of coefficients of the final slack form and let $y_i = -c_{n+i}$ for $(n+i) \in N$; 0 otherwise.
- Under these conditions:
 - x is the optimal solution for the primal problem
 - y is the optimal solution for the dual problem
 - and,

$$\sum_{j=1}^n c_j x_j = \sum_{i=1}^m b_i y_i$$

Fundamental Theorem in LP

- Any problem L in Standard Form either:
 - Has an Optimal Solution with finite value
 - Is Infeasible
 - Is Unbounded
 - If L is Infeasible, the Simplex algorithm returns “infeasible”
 - If L is Unbounded, the Simplex algorithm returns “unbounded”
 - Otherwise the Simplex Algorithm returns an Optimal Solution with a finite objective function value

Summary

- Motivational Problem
- Canonical Forms: Standard and Slack
- Problem Formulation
- The Simplex Algorithm
- Feasible Initial Solutions
- Duality and Correctness of Simplex